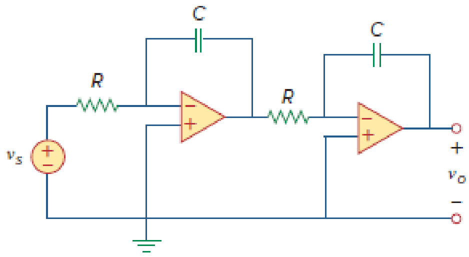
Chapter 16, Problem 98P

Problem

Determine whether the op amp circuit in Fig. 16.107 is stable.

Figure 16.107:



Step-by-step solution

Step 1 of 1

Let V_{O1} be the voltage at the out put of first Op - Amp

$$\frac{V_{O1}}{V_S} = \frac{\frac{-1}{SC}}{R} = \frac{-1}{SCR} \qquad \frac{V_O}{V_{O1}} = \frac{-1}{SRC}$$

$$H(S) = \frac{V_O}{V_S} = \frac{1}{S^2R^2C^2}$$

$$h(t) = \frac{t}{R^2C^2}$$

$\lim_{t\to\infty} h(t) = \infty$ i.e Output is unbunded

hence , circuit is unstable